**Example 3.6** Find the length of an arc of a circle of radius $5\ cm$ subtending a central angle measuring $15°$.

**Solution:**
Let $s$ be the length of the arc of a circle of radius $r$ subtending a central angle $\theta$.
Then $s = r\theta$.

We have, $\theta = 15° = 15 \times \frac{\pi}{180} = \frac{\pi}{12}$ radians

So that, $s = r\theta$ gives $s = 5 \times \frac{\pi}{12} = \frac{5\pi}{12}\ cm$

In the product $r\theta$ , $\theta$ must always be in radians.

**Example 3.7** If the arcs of same lengths in two circles subtend central angles $30°$ and $80°$, find the ratio of their radii.

**Solution:**
Let $r_1$ and $r_2$ be the radii of the two given circles and $l$ be the length of the arc.

$\theta_1 = 30° = \frac{\pi}{6}$ radians

$\theta_2 = 80° = \frac{4\pi}{9}$ radians

Given that $l = r_1\theta = r_2\theta$

Thus, $\frac{\pi}{6}\, r_1 = \frac{4\pi}{9}\, r_2$

$\frac{r_1}{r_2} = \frac{8}{3}$ which implies $r_1 : r_2 = 8 : 3$.

## Exercise - 3.2

1. Express each of the following angles in radian measure:
   (i) $30°$ (ii) $135°$ (iii) $-205°$ (iv) $150°$ (v) $330°$.
2. Find the degree measure corresponding to the following radian measures
   (i) $\frac{\pi}{3}$ (ii) $\frac{\pi}{9}$ (iii) $\frac{2\pi}{5}$ (iv) $\frac{7\pi}{3}$ (v) $\frac{10\pi}{9}$.
3. What must be the radius of a circular running path, around which an athlete must run $5$ times in order to describe $1\ km$?
4. In a circle of diameter $40\ cm$, a chord is of length $20\ cm$. Find the length of the minor arc of the chord.
5. Find the degree measure of the angle subtended at the centre of circle of radius $100\ cm$ by an arc of length $22\ cm$.
6. What is the length of the arc intercepted by a central angle of measure $41°$ in a circle of radius $10\ ft$?
7. If in two circles, arcs of the same length subtend angles $60°$ and $75°$ at the centre, find the ratio of their radii.